

## Your PC's downfall—its software

You are as vulnerable as the software that's installed on your machine. As soon as you install Linux, be sure to update all the packages you installed. This is especially important if you've installed an older version of Linux—several security issues would have arisen since its release, and addressed in subsequent updates.

By default, Linux tends to start up a whole bunch of unwanted services, and

Each service listed has a checkbox next to it; if this box is ticked, the service will automatically start when the machine boots. Uncheck the box to prevent the service from automatically starting.

### What you don't need

Here's a list of services that you might consider switching off:

#### APMD (Advanced Power Management

**Daemon)**—Only needed if your power source is a battery, as in laptops.

**atd (AT Daemon)**—Used to start a job like cron does; it's not needed, as you can set the system to run scheduled jobs using cron.

**cups and lpd**—Only needed if you have a printer connected.

**httpd (the Web server service)**—Only needed if you run a Web server.

**Sendmail (the mail transport agent)**—Only needed if you run a mail server.

Chandrasekar points out that people routinely run Sendmail on Linux or Unix, and it doesn't take much to attack known vulnerabilities in Sendmail. Worse, people can exploit your computer by running port scans and using your computer as a mail relay to

send spam.

These are just a few services that you can terminate for good, but there will always be even more that you're left in doubt about. Here's where descriptions come in handy. Highlight a service and read through its description on the right. For instance, telnet and sshd are both used to access the machine from

## Linux at IIT-B

With huge numbers of computers scattered all over campus—over 1,800 PCs connecting just from the 13 hostels—IIT Bombay's networks have Windows and Linux-based PCs connected to their network backbone.

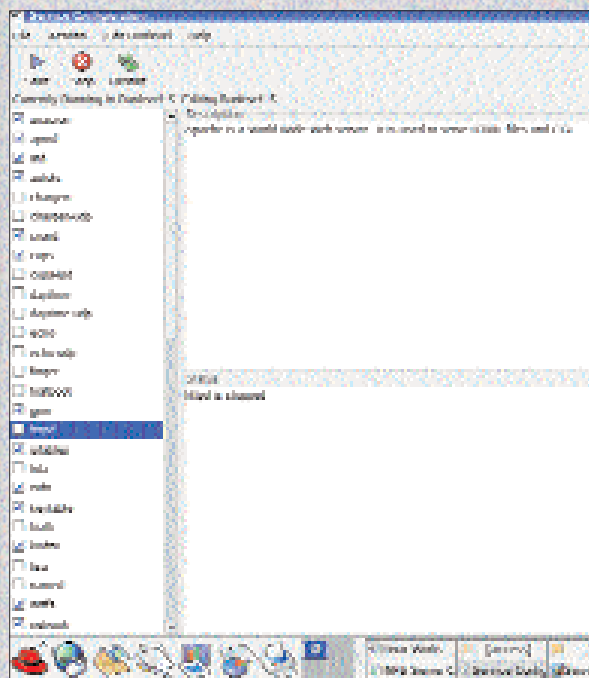
We spoke to Dr J P Maharana, the systems manager at the Computer Center, who told us that IIT-Bombay has started recommending using Linux-based PCs, ever since the Blaster attack last year bogged down their networks—due to infected Windows PCs.

Dr. Maharana himself runs Linux on his workstation and says that just about everything can be done with Linux. His advice to both Linux and Windows users is, "Get all critical system updates, and ensure that your antivirus definitions are updated as well."

remote locations, so if you don't plan on doing so, turn them off—this will prevent others from doing it.

Although you're safest with a minimum of services or daemons running, make sure you have every service that's vital to the functioning of your machine turned on. To list out the whole set of services, type in '**chkconfig -list**'. This lists out all services, including those started by xinetd—a daemon that starts network services.

Chkconfig will also tell you which services start on which run-level. Think of run-levels as different modes of operating

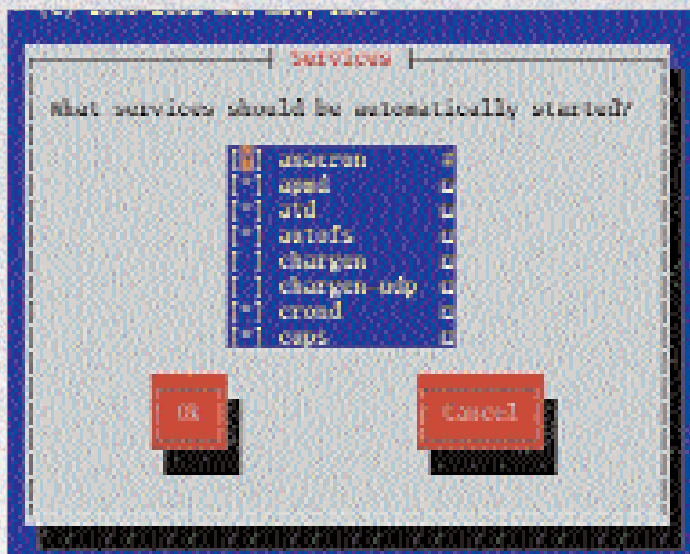


The Service Configuration applet lets you start, stop and restart services

what's worse, having them running makes your computer more vulnerable to attacks. First, look for daemons—system services that are listed. These wait for a signal or connection and make a list of what you don't want or need. For instance, the Common Unix Printer System (cups) and Linux Printer Daemon (lpd) daemons are useless if you don't have a printer connected, either directly or via a network.

To look at all the services currently running, click on the Red Hat button and go to **System Settings > Server Settings > Services**. This opens up the Service Configuration applet, which lists out all the daemons that can run on the machine.

The left-hand pane lists out the services, and the description and status are listed on the right. To start or shut down a service, highlight it and select Start, or Stop, from the menu bar on top.



ntsysv is almost as good as chkconfig, and is easier to work with because of the Curses interface